

Introduction to Contract Theory and Auctions
Lecture 5
Screening: Simple Examples

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ECON 403/503
Fall 2021

- A contracting problem with **Hidden Information**
- Uninformed party (principal) offers contract to informed party (agent)
- Examples
 - Insurance
 - Insuree knows her risk, insurer does not
 - Insurer offers several packages with different premiums and deductibles
 - Finance
 - Borrower knows the risk of project, lender does not
 - Lender offers several packages with different interest rates and collateral requirements
 - Hiring
 - Applicants know their ability, employer does not
 - Employer offers different packages varying wages, bonuses, etc.
 - Pricing
 - Buyer knows her valuation of the product, seller does not
 - Seller offers different qualities at different prices, or quantity discounts

The Principal-Agent Model: Delegation

- The problem arises in particular when a **principal** *delegates* a task to an **agent**.
- Delegation can be motivated by
 - The possibility of benefiting from increasing returns associated with the division of tasks.
 - Principal's lack of ability or time to perform the task himself.
 - Bounded rationality or complexity issues that the principal faces.
- Because of delegation, the agent may get access to **information** not available to the principal.
 - The opportunity cost of the task.
 - The technology.
 - The ability of the agent to perform the task.
- This information is **private knowledge** of the agent, i.e. there is **adverse selection**

- Both private and public transactions are subject to adverse selection.
 - Landlord delegates cultivation of his land to a tenant, who will be the only one observing the exact weather conditions.
 - A client delegates his defense to an attorney, who is the only one that knows the true difficulty of the case.
 - An investor delegates the management of his portfolio to a broker, who privately knows the prospects of the possible investments.
 - Stockholders delegate the firm's day-to-day decisions to a manager, who is the only one who observes the precise business conditions.
 - The Department of Defense procures a good from the military industry without knowing its exact cost structure.
 - A regulator contracts for service with a Public Utility company without having complete information about its technology

- The information gap between principal and agent has fundamental implications for the design of the bilateral contract that they sign.
- The contract must **elicit** the private information.
- However, this can be done only by giving up some **information rent** to the privately informed agent.
- This information rent distorts the volume of trade achieved under asymmetric information.
- There is an Allocative Efficiency-information rent trade off.
- The first best contract, the one that achieves allocative efficiency is not feasible.
- We will look for second best contracts, the ones that find the optimal efficiency-information rent trade off.

A basic model of hidden information

- One-shot relationship between the principal and the agent.
 - The parties cannot rely on repetition to achieve efficient trades:
 - Cooperation through repetition.
 - Learning private information through repetition.
 - Reputation effects.
- The short-term relationship implies that the relationship can only be regulated by a **contract**.
- We implicitly assume that there exists a legal framework, a benevolent court of law, which can enforce the contractual relationship.
 - This feature prevails in most studies of market economies.
 - If you want to see a model in which only power and coercion regulate the exchange of resources, read Piccione & Rubinstein (2006).

Screening by a Monopolist: A stylized example

- You are a wine producer.
- A fraction c of your customers are connoisseurs while the rest are novice.
- You can produce a high quality wine and/or a low quality wine.
- The following table gives the cost of producing the two qualities of wine as well as the willingness to pay for them by the two types of customers.

	Cost	Willingness to pay	
		Connoisseur	Novice
Low Quality	2	10	6
High Quality	10	30	12

- You choose which qualities to produce and their prices to maximize your expected profit.

- Suppose you can tell your customer's type.
- What should you do?
- Produce both qualities.
- Sell high quality to connoisseur at \$30 and low quality to novice at \$6.
- Your profit is

$$(30 - 10) \times c + (6 - 2) \times (1 - c) = 20c + 4(1 - c)$$

- Maximizes the total surplus:
 - Extra benefit from high quality is bigger (smaller) than the extra cost for connoisseur (novice)
- This is first degree (or perfect) price discrimination.
- It is great if you can implement it.
- But usually you cannot observe types.
- You may try to ask them their types. What would they tell you?

You cannot observe types

- What happens if you offer high quality at \$30 and low quality at \$6?
- Here are the net payoffs of the two types from purchasing the two qualities

	Net Payoffs	
	Connoisseur	Novice
Low Quality	$10 - 6 = 4$	$6 - 6 = 0$
High Quality	$30 - 30 = 0$	$12 - 30 = -18$

- High type wants to buy low quality.
- Everybody will buy low quality.
- Your profit will be $6 - 2 = 4$ only.
- Less than if you could observe types and perfectly discriminate.
- What is the best you can do in this case?

You cannot observe types

There are several things that you can do:

- 1 Offer only low quality wine.
- 2 Offer only high quality wine.
- 3 Offer both so that Connoisseur (C) chooses high and Novice (N) chooses low quality.

Offer only low Quality

	Cost	Willingness to pay	
		Connoisseur	Novice
Low Quality	2	10	6
High Quality	10	30	12

- What price should you set?
- If you set a price at \$6 both types buy.
 - Your profit is $6 - 2 = 4$.
- If you set price at \$10 only type C buys.
 - Your profit is $c(10 - 2) = 8c$.
- Therefore, your best pricing policy and resulting profit is

	Price	Profit
$c \leq 1/2$	6	4
$c > 1/2$	10	$8c$

	Cost	Willingness to pay	
		Connoisseur	Novice
Low Quality	2	10	6
High Quality	10	30	12

- What price should you set?
- If you set a price at \$12 both types buy.
 - Your profit is $12 - 10 = 2$.
- If you set price at \$30 only type C buys.
 - Your profit is $c(30 - 10) = 20c$.
- Therefore, your best pricing policy and resulting profit is

	Price	Profit
$c \leq 1/10$	12	2
$c > 1/10$	30	$20c$

You offer both qualities

- You want C to buy high and N to buy low quality wine.
- You want to choose P_H and P_L to maximize your expected profits

$$c(P_H - 10) + (1 - c)(P_L - 2)$$

- C must prefer high to low quality

$$30 - P_H \geq 10 - P_L \quad (IC_C)$$

- N must prefer low quality

$$6 - P_L \geq 12 - P_H \quad (IC_N)$$

- The above are called **Incentive Compatibility** (IC) constraints.
- And you want them to prefer to buy

$$30 - P_H \geq 0 \quad (IR_C)$$

$$6 - P_L \geq 0 \quad (IR_N)$$

- The above are called **Individual Rationality** (IR) (or participation) constraints.

Solving the problem

- ① (IR_N) holds with equality $P_L = 6$.

Suppose instead that $6 - P_L > 0$. Then

$$30 - P_H \geq 10 - P_L > 6 - P_L > 0$$

- We could increase P_H and P_L by the same amount without violating IC constraints and increase our profit.
- This also shows that we can neglect (IR_C) .

- ② (IC_C) holds with equality: $30 - P_H = 10 - P_L$.

Suppose instead that $30 - P_H > 10 - P_L > 6 - P_L = 0$. Then

$$30 - P_H > 10 - P_L > 6 - P_L = 0.$$

- We could increase P_H without violating any constraints and increase our profit.

- ③ Therefore, we have

$$P_L = 6, P_H = P_L + (30 - 10) = 30 - (10 - 6) = 26.$$

You offer both quantities

- So prices are $P_L = 6$ and $P_H = 26$.
- Notice that you extract the full surplus from the Novice but leave \$4 surplus to the Connoisseur.
- Why?
- Remember what happens if you set prices at perfect information levels, i.e., $P_L = 6$ and $P_H = 30$.
 - Novice will happily buy this low quality of wine.
 - But, Connoisseurs want to buy low quality as well!
- The amount of surplus you leave to Connoisseur is the minimum you have to give her to buy high quality
 - This is called **information rent**.
- What is your profit?

$$(26 - 10) \times c + (6 - 2) \times (1 - c) = 16c + 4(1 - c)$$

- Compare this with your profit under observable types.

$$(30 - 10) \times c + (6 - 2) \times (1 - c) = 20c + 4(1 - c)$$

What is the best?

	Low Only		High Only		Both	
	Price	Profit	Price	Profit	Price	Profit
$c \leq 1/10$	6	4	12	2	(26, 6)	$16c + 4(1 - c)$
$1/10 < c < 1/2$	6	4	30	$20c$	(26, 6)	$16c + 4(1 - c)$
$c \geq 1/2$	10	$8c$	30	$20c$	(26, 6)	$16c + 4(1 - c)$

- Both is always better than Low only.
- If $c \leq 1/2$ producing both is best.
- If $c > 1/2$ producing High only is the best.
- Getting surplus from Novice customers is costly.
 - You have to leave the Connoisseurs some surplus.
 - As the fraction of Connoisseurs increases the benefit decreases and the cost increases.
- Therefore, if there are enough Connoisseurs around you don't care much about getting the surplus from the Novice.

What happens if you can determine the level of quality of wine?

- You are a wine producer.
- Half the customers have High taste and half have Low taste.
- If a customer buys q quality wine and pays t her payoff is
 - High taste $u_H = 3q - t$
 - Low taste $u_L = 2q - t$
- If she does not buy her payoff is zero.
- Cost of producing quality q is q^2 .
- You choose q and t to maximize your profit $t - q^2$.

If you can observe the types you get the highest possible profits

- Produce two qualities: $q_H, q_L \geq 0$ and set prices t_L, t_H to maximize

$$\frac{1}{2}(t_L - q_L^2) + \frac{1}{2}(t_H - q_H^2)$$

subject to

$$3q_H - t_H \geq 0 \quad 2q_L - t_L \geq 0$$

- Since your profit is increasing in t_L and t_H , you should set $t_H = 3q_H$, $t_L = 2q_L$.
- Substitute into the objective function and the problem becomes: Choose $q_H, q_L \geq 0$ to maximize:

$$\frac{1}{2}(2q_L - q_L^2) + \frac{1}{2}(3q_H - q_H^2)$$

- Choose $q_H, q_L \geq 0$ to maximize:

$$\frac{1}{2}(2q_L - q_L^2) + \frac{1}{2}(3q_H - q_H^2)$$

- If at the solution, $q_H, q_L > 0$, then the derivative of this function with respect to q_L and q_H must be zero:

$$\frac{1}{2}(2 - 2q_L) = 0$$

$$\frac{1}{2}(3 - 2q_H) = 0$$

which are solved as $q_L = 1$ and $q_H = 3/2$.

- Therefore,

$$t_L = 2q_L = 2 \quad \text{and} \quad t_H = 3q_H = \frac{9}{2}$$

- Your profit is

$$\frac{1}{2} \left(\frac{9}{2} - \frac{9}{4} \right) + \frac{1}{2}(2 - 1) = \frac{13}{8}$$

If you cannot observe types

- What happens if you offer quality $3/2$ at price $9/2$ and quality 1 at price 2?
- High type's payoff if she buys low quality is higher than buying high quality

$$3 \times 1 - 2 = 1 > 0$$

- Everybody will buy low quality
- Your profit will be at most $2 - 1 = 1$
- Less than if you could observe types and perfectly discriminate
- What is the best you can do in this case?
- Offer a high quality wine at high price and a low quality wine at low price so that
 - High taste customer buys the high quality wine
 - Low taste customer buys the low quality wine
 - You maximize your expected profit

You want to choose (t_H, q_H) and (t_L, q_L) to maximize

$$\frac{1}{2}(t_L - q_L^2) + \frac{1}{2}(t_H - q_H^2)$$

But you also want

- High taste customer buys the high quality wine

$$3q_H - t_H \geq 3q_L - t_L \quad (IC_H)$$

- Low taste customer buys the low quality wine

$$2q_L - t_L \geq 2q_H - t_H \quad (IC_L)$$

The above are called **Incentive Compatibility** (IC) constraints

- And you want them to buy

$$3q_H - t_H \geq 0 \quad (IR_H)$$

$$2q_L - t_L \geq 0 \quad (IR_L)$$

Called **Individual Rationality** (IR) constraints.

Solving the problem

- ① (IR_L) holds with equality. Suppose instead that $2q_L - t_L > 0$. Then

$$3q_H - t_H \geq 3q_L - t_L \geq 2q_L - t_L > 0$$

- We could increase t_H and t_L by the same amount without violating IC constraints and increase our profit.
- This also shows that we can neglect (IR_H) .

- ② (IC_H) holds with equality. Suppose instead that $3q_H - t_H > 3q_L - t_L$. Then

$$3q_H - t_H > 3q_L - t_L \geq 2q_L - t_L = 0$$

- We could increase t_H without violating any constraints and increase our profit.

- ③ $q_H \geq q_L$. Because from (IC_L) and (IC_H) we have

$$2(q_H - q_L) \leq t_H - t_L \leq 3(q_H - q_L).$$

- ④ We can neglect (IC_L) because

$$2(q_H - q_L) \leq 3(q_H - q_L) = t_H - t_L.$$

- Therefore, Low type's individual rationality and High type's incentive compatibility constraints hold as equalities and we can ignore the other two constraints
- So, the problem becomes: Choose (t_H, q_H) and (t_L, q_L) to maximize

$$\frac{1}{2}(t_H - q_H^2) + \frac{1}{2}(t_L - q_L^2)$$

subject to

$$t_L = 2q_L$$

$$3q_H - t_H = 3q_L - t_L$$

- Substituting the constraints into the profit function yields

$$\frac{1}{2}(3q_H - q_L - q_H^2) + \frac{1}{2}(2q_L - q_L^2)$$

- The problem therefore becomes: Choose $q_L, q_H \geq 0$ to maximize

$$\frac{1}{2}(3q_H - q_L - q_H^2) + \frac{1}{2}(2q_L - q_L^2)$$

- If the solution has $q_H, q_L > 0$, then first derivatives with respect to q_H and q_L must be equal to zero

$$\begin{aligned}\frac{1}{2}(3 - 2q_H) &= 0 \\ \frac{1}{2}(-1) + \frac{1}{2}(2 - 2q_L) &= 0\end{aligned}$$

- These are solved as

$$q_H = 3/2 \quad \text{and} \quad q_L = 1/2$$

- We can use the two constraints to solve for t_L, t_H :

$$\begin{aligned}t_L &= 2q_L \Rightarrow t_L = 1 \\ 3q_H - t_H &= 3q_L - t_L \Rightarrow t_H = 4\end{aligned}$$

Optimal wine qualities

Your optimal quality/price choice:

- Quality $3/2$ wine at price 4.
- Quality $1/2$ wine at price 1.

Remember that if you could observe types you get

- Quality $3/2$ wine at price $9/2$.
- Quality 1 wine at price 2.

The problem with this was that high taste customer preferred buying the low quality wine.

What do you do to solve the problem?

- Keep giving him the same quality at lower price.
 - This gives him positive surplus.
- Reduce quality and price for the low taste customer.
 - You do this to keep high type's surplus at minimum.
- Low taste customer was not a problem.
 - Keep his surplus at zero.

- Although this is a very simple example, several results carry over to more general models.
- Low type's individual rationality constraint binds: she receives zero surplus.
- High type's incentive compatibility constraint binds.
- As a result, High type obtains positive surplus.
 - Known as **information rent**.
 - You leave rent to High type to dissuade her from mimicking the Low type.
- High type consumes the efficient quantity.
 - This is the quantity that maximizes total surplus: revenue – cost

$$\text{marginal revenue} = 3 = 2q_H = \text{marginal cost}$$

- Low type consumes less than the efficient amount.